Technical Report: H3-Based Spatial Representation and Relationship Modelling for Heterogeneous Hierarchies

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Abstract

This technical report provides the complete algorithmic specifications supporting the research paper "H3-Based Computation of Qualitative Spatial Relationships Across Heterogeneous Geographic Hierarchies" (Satoti and Abdelmoty, submitted to *Transactions in GIS*). It documents: (1) the adaptive H3 cell assignment algorithm with full pseudocode, mathematical framework, and resolution mapping thresholds; (2) the spatial relationship classification algorithm with pseudocode for all 17 relationship types and 6 auxiliary functions; and (3) representative validation examples from the Welsh dataset. This report ensures methodological transparency and reproducibility.

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1 Introduction

Traditional geographic information systems (GIS) compute spatial relationships through vector geometry operations, classifying spatially separated features as simply `DISJOINT` with no mechanism for graded proximity or hierarchical interactions across administrative divisions. The associated research paper presents a methodology using Uber's H3 Discrete Global Grid System (DGGS) as a computational bridge between heterogeneous geographic hierarchies (administrative, electoral, and postal).

This technical report provides the detailed algorithmic specifications that underpin the methodology. It complements the research paper by including the full pseudocode for both core algorithms and all auxiliary functions, enabling independent reimplementation. For the

theoretical motivation, related work, and empirical analysis, the reader is referred to the research paper.

2 Algorithm 1: Adaptive H3 Cell Assignment

This algorithm transforms irregular spatial units into standardised hexagonal representations by assigning adaptive H3 resolutions based on feature area, applying geometry buffering to ensure boundary coverage, and generating H3 cell sets via polyfill.

2.1 Mathematical Framework

Resolution Mapping Function. A threshold-based function $T : \mathbb{R}^+ \to \{0, 1, \ldots, 15\}$ assigns H3 resolutions based on the area A_i (in m^2) of spatial unit i. The thresholds are calibrated so that the assigned resolution produces cells whose average area is substantially smaller than the spatial unit being represented. The complete mapping is provided in Table 1.

Geometry Buffering. A minimal buffer is applied in a metre-based CRS (EPSG:27700, British National Grid):

$$d_r = 0.5 \times \text{h3.edge_length}(r, \text{unit='m'})$$

The factor 0.5 ensures that H3 cell centres near the geometry's boundary are included in the polyfill operation. The buffered geometry is then reprojected to WGS84 (EPSG:4326) for H3 polyfill.

Fallback Mechanism. If the initial resolution yields no H3 cells, the resolution is incrementally increased:

$$r_i^{(k)} = T(A_i) + k, \quad k = 0, 1, 2, 3$$

The process terminates when cells are generated or the maximum of three increments is reached. The buffer is recalculated at each fallback resolution.

2.2 Resolution Mapping Thresholds

Table 1: Complete Adaptive H3 Resolution Mapping Function $T(A_i)$

Resolution (r_i)	**Area Threshold (A_i in m^2)**	**Avg. Cell Area (m^2)**
0	$A_i \geq$ 609,788,441,794,100	4,357,449,416,078
1	86,801,780,399,000 $\leq A_i <$ 609,788,441,794,100	609,788,441,794
2	12,393,434,655,100 $\leq A_i <$ 86,801,780,399,000	86,801,780,399
3	1,000,000,000,000 $\leq A_i <$ 12,393,434,655,100	12,393,434,655
4	100,000,000,000 $\leq A_i <$ 1,000,000,000,000	1,770,347,655
5	10,000,000,000 $\leq A_i <$ 100,000,000,000	252,903,858
6	1,000,000,000 $\leq A_i <$ 10,000,000,000	36,129,062
7	100,000,000 $\leq A_i <$ 1,000,000,000	5,161,293
8	10,000,000 $\leq A_i <$ 100,000,000	737,328
9	1,000,000 $\leq A_i <$ 10,000,000	105,333
10	500,000 $\leq A_i <$ 1,000,000	15,048
11	100,000 $\leq A_i <$ 500,000	2,150
12	10,000 $\leq A_i <$ 100,000	307
13	1,000 $\leq A_i <$ 10,000	44
14	100 $\leq A_i <$ 1,000	6
15	$A_i <$ 100	0.9

Note: Individual H3 cells vary slightly in area depending on their position on the icosahedral grid; the values reported here represent global averages from the H3 specification. In the Welsh

dataset, assigned resolutions ranged from 5 (Wales, ≈21,218 km^2) to 11 (small postal sectors, ≈0.1–0.5 km^2), with 86.3% of features receiving resolutions 8 or 9.

2.3 Pseudocode

Algorithm 1 H3 Cell Assignment for Spatial Units

```
Require: A dictionary unit_dict mapping Spatial_unit_ID to Unit objects with attributes hierarchy_id,
    unit_level, geometry, name, and a threshold-based resolution mapping function T(A)
Ensure: A list of records containing Spatial_unit_ID, assigned_res, h3_cells
records ← EMPTY_LIST
for hid in UNIQUE(unit_dict.values.hierarchy_id) do
    hierarchy_units ← FILTER(unit_dict.values, hierarchy_id = hid)
    for level in UNIQUE(hierarchy_units.unit_level) do
        level_units ← FILTER(hierarchy_units, unit_level = level)
        for unit in level_units do
            projected_geometry ← PROJECT(unit.geometry, EPSG:27700)
            area ← COMPUTE_AREA(projected_geometry)                          ▷ area in m^2
            if area ≤ 0 then
                APPEND({ID: unit.ID, res: −1, cells: ∅}, records)
                continue
            end if
            res ← T(area)
            std_geom ← STANDARDIZE(projected_geometry)      ▷ fix self-intersections, dissolve multipart
            buf_dist ← 0.5· h3.edge_length(res, unit='m')
            buf_geom ← BUFFER(std_geom, buf_dist)
            h3_geom ← PROJECT(buf_geom, EPSG:4326)                   ▷ reproject to WGS84
            h3_cells ← H3_POLYFILL(h3_geom, res)
            attempts ← 0
            while h3_cells = ∅ and attempts < 3 do
                res ← res +1
                buf_dist ← 0.5· h3.edge_length(res, unit='m')
                buf_geom ← BUFFER(std_geom, buf_dist)
                h3_geom ← PROJECT(buf_geom, EPSG:4326)
                h3_cells ← H3_POLYFILL(h3_geom, res)
                attempts ← attempts +1
            end while
            h3_cells ← UNIQUE(h3_cells)                                      ▷ deduplicate
            APPEND({ID: unit.ID, res: res, cells: h3_cells}, records)
        end for
    end for
end forreturn records
```

3 Algorithm 2: Spatial Relationship Computation

This algorithm classifies the spatial relationship between two features based on their H3 cell sets, resolutions, and hierarchical context. It produces one of 17 relationship labels grouped into topological, proximity, and hierarchical categories.

3.1 Metric Logic

The overlap ratio ρ characterises the degree of intersection:

$$\rho = \frac{|A \cap B|}{\min(|A|, |B|)}, \quad 0 \leq \rho \leq 1 \tag{1}$$

Proximity is measured by the minimum H3 grid distance:

$$d(A, B) = \min_{a \in A,\, b \in B} \texttt{gridDistance}(a, b) \tag{2}$$

3.2 Main Relationship Computation Algorithm

Algorithm 2: Compute Spatial Relationship Between Two Features

Input: H3 cell sets $\mathcal{C}_i$, $\mathcal{C}_j$; resolutions r_i, r_j

Output: Relationship label

```
C_i ← SET(C_i), C_j ← SET(C_j)
if C_i = ∅ or C_j = ∅ then
    return Error: Not Valid
end if
if r_i = r_j then                                                    ▷ Same resolution
    if C_i ⊆ C_j and C_i ≠ C_j then
        return Complete Containment
    else if C_j ⊆ C_i and C_j ≠ C_i then
        return Complete Containment
    end if
    if C_i = C_j then
        return Identical
    end if
    S ← C_i ∩ C_j
    if S ≠ ∅ then
        if |S|/ min(|C_i|, |C_j|) ≤ 0.05 and AreSharedCellsOnBoundary(C_i, C_j, S) then
            return Touch
        else
            ρ ← |S|/ min(|C_i|, |C_j|)
            if ρ ≤ 0.10 then return Intersect
            else return Overlap
            end if
        end if
    end if
    if AdjacentCells(C_i, C_j) then
        return Neighbour
    end if
    d ← min_{a∈C_i, b∈C_j} gridDistance(a, b)
    if d = 2 then return Close
    else if d ∈ {3, 4} then return Near
    else if d > 4 then return Far
    else return Far Away
    end if
else                                                  ▷ Different resolution — hierarchical
    r_c ← min(r_i, r_j); r_f ← max(r_i, r_j)
    C_f ← cell set at finer resolution; C_c ← cell set at coarser
    C_f^↑ ← ClimbParents(C_f, r_c)
    if C_f^↑ ⊆ C_c then
        if r_c = r_f − 1 then
            if AllGeometriesContained(C_f, C_c) then
                return Direct Parent, Complete Containment
            else return Direct Parent, Partial Containment
            end if
        else
            if AllGeometriesContained(C_f, C_c) then
                return Ancestor, Complete Containment
            else return Ancestor, Partial Containment
            end if
        end if
    else if SomeCellsHaveParents(C_f^↑, C_c) then
        S ← C_f^↑ ∩ C_c
        B_c ← GetBorderCells(C_c)
        if S ⊆ B_c then return Hierarchical Touch
        else return Hierarchical Overlap
        end if
    else
        return Disjoint
```

```
    end if
end if
```

3.3 Auxiliary Functions

Algorithm 2 ClimbParents: Elevate Cells to Coarser Resolution

Require: H3 cell set $\mathcal{C}_f$, target resolution r_c
Ensure: Coarsened cell set $\mathcal{C}_f^{\uparrow}$

```
C_f^↑ ← ∅
for all c ∈ C_f do
    p ← cellToParent(c, r_c)
    C_f^↑ ← C_f^↑ ∪ {p}
end for
return C_f^↑
```

Algorithm 3 AllGeometriesContained: Geometric Containment Check

Require: H3 cell sets $\mathcal{C}_f$, $\mathcal{C}_c$
Ensure: `true` if $\bigcup_{c \in \mathcal{C}_f} \texttt{geometry}(c) \subseteq \bigcup_{c \in \mathcal{C}_c} \texttt{geometry}(c)$

```
G_f ← ⋃_{c∈C_f} geometry(c)
G_c ← ⋃_{c∈C_c} geometry(c)
if G_c ⊇ G_f then return true
else return false
end if
```

Algorithm 4 AdjacentCells: Check Cell Adjacency

Require: H3 cell sets $\mathcal{C}_i$, $\mathcal{C}_j$
Ensure: `true` if any cell in $\mathcal{C}_i$ is adjacent to any cell in $\mathcal{C}_j$

```
for all a ∈ C_i do
    N_a ← neighbors(a)
    if N_a ∩ C_j ≠ ∅ then return true
    end if
end for
return false
```

Algorithm 5 AreSharedCellsOnBoundary: Boundary Cell Check

Require: H3 cell sets $\mathcal{C}_i$, $\mathcal{C}_j$, shared cells S
Ensure: `true` if all shared cells in S are boundary-adjacent to both features

```
for all s ∈ S do
    N_s ← neighbors(s)
    N_i ← N_s ∩ (C_i \ S)
    N_j ← N_s ∩ (C_j \ S)
    if N_i = ∅ or N_j = ∅ then return false
    end if
end for
return true
```

Algorithm 6 SomeCellsHaveParents: Partial Hierarchical Match

Require: H3 cell sets $\mathcal{C}_f^{\uparrow}$, $\mathcal{C}_c$
Ensure: `true` if $\mathcal{C}_f^{\uparrow} \cap \mathcal{C}_c \neq \emptyset$

```
if C_f^↑ ∩ C_c ≠ ∅ then return true
else return false
end if
```

Algorithm 7 GetBorderCells: Identify Border Cells

Require: H3 cell set $\mathcal{C}$
Ensure: Set of border cells $\mathcal{B}$

```
B ← ∅
for all c ∈ C do
    N_c ← neighbors(c)
    if N_c ⊈ C then
        B ← B ∪ {c}
    end if
end for
return B
```

4 Evaluation Summary

The framework was evaluated on 3,242 spatial units across Welsh administrative, electoral, and postal hierarchies, comprising 2,627,499 cross-hierarchy feature pairs.

Table 2: Summary of Key Results

Metric	**Result**
Total feature pairs computed	2,627,499
H3 computation time	49 minutes
DE-9IM computation time	177 minutes
Computational speedup	3.6×
Mean area approximation error	8.30% (SD = 4.80%)
Identical relationships (H3)	433
EQUALS relationships (DE-9IM)	11
Hierarchical relationships detected	6,537
Proximity categories (non-disjoint)	1,025,984

5 Data and Code Availability

The Welsh administrative, electoral, and postal boundary datasets used in this study are derived from Ordnance Survey Open Data (`https://osdatahub.os.uk/downloads/open`). The algorithmic specifications and validation examples are available at: `https://github.com/satoti/H3-Computation-Spatial-Relationships-Across-Heterogeneous-Hierarchies`.

References

1. Satoti, A. and Abdelmoty, A. I. (submitted). H3-Based Computation of Qualitative Spatial Relationships Across Heterogeneous Geographic Hierarchies. Manuscript submitted to *Transactions in GIS*.

2. Satoti, A. and Abdelmoty, A. I. (2025). A GIS-Native Framework for Qualitative Place Models: Implementation and Evaluation. *ISPRS International Journal of Geo-Information*, 14(12), 474. `https://doi.org/10.3390/ijgi14120474`.

3. Uber Technologies. H3: Hexagonal Hierarchical Geospatial Indexing System. `https://h3geo.org/docs/core-library/restable/`.